Blockchain-Enabled

Quantum-Resistant Healthcare Systems

Integrating LAAP · NIST PQC (FIPS 203/204) · DP-FL · Blockchain

Presented By: Hridiyansh Shukla | Roll No.: 2427030591

Project Guide: Dr. Usha Jain

Department: CSE · School of CSE · Manipal University Jaipur

Course: Project Based Learning 2 · Jan – May 2026

Introduction: The IoMT Healthcare Revolution

Why current architectures are no longer sufficient

$260B

Global IoMT Market by 2027

79M

Patient records exposed — Anthem 2015 Breach

2017 WannaCry

Ransomware paralysed the entire UK NHS

I — IoMT

Wearables, implantables & remote sensors generating continuous patient telemetry in real-time

T — Threat

Centralized EHR databases create single points of failure — one breach = millions exposed

N — Need

Decentralized Blockchain + IPFS architecture eliminates central attack vectors

G — Gap

IoMT devices are resource-constrained — classical crypto (RSA/ECC) is simply too heavy to run

Q — Quantum

Shor's algorithm on a quantum computer will break all deployed RSA & ECC — harvest-now attacks are live today

Problem Statement: A Three-Layer Security Crisis

Converging threats demanding a unified architectural response

Device Resource Constraints

- IoMT MCUs: 8–32 bit, ≤512 KB RAM, battery-limited
- RSA-3072 auth latency: >2,000 ms on embedded hardware
- Classical crypto exhausts CPU & battery before clinical use
- $O(k^3)$ modular exponentiation — computationally prohibitive

71.96% overhead reduction needed

Quantum Computing Threat

- Shor's Algorithm breaks RSA & ECC in polynomial time
- "Harvest Now, Decrypt Later" — EHR archives at risk today
- Genetic & biometric data: lifelong sensitivity
- CRQC projection: feasible within the next decade

100% of deployed PKI vulnerable to CRQC

Centralized AI & Privacy Risk

- Central ML for IDS requires raw patient data aggregation
- Violates GDPR data minimization & HIPAA minimum-necessary
- Centralized models vulnerable to poisoning & inference attacks
- No formal correctness guarantee on smart contract logic

95.48% target IDS accuracy with privacy

Literature Review & Research Gaps

Evolutionary analysis: Classical crypto → Lightweight → Post-Quantum

Primitive	Hard Problem	Key Size	Time Complexity	Quantum Safe?	IoMT Suitability
RSA-3072	Integer Factorization	3072 bits	$O(k^3)$	NO ✗	Not Suitable
ECC (P-256)	EC Discrete Log	256 bits	$O(k^2)$	NO ✗	Moderate
LAAP (Curve25519)	EC Discrete Log	256 bits	$O(k^2)$ / O(1)*	NO ✗	Highly Suitable
ML-KEM (FIPS 203)	Module-LWE	800 B (pk)	O(n log n) NTT	YES ✓	Gateway/WAN
ML-DSA (FIPS 204)	Module-SIS	1312 B (pk)	O(n log n) NTT	YES ✓	Gateway/WAN
TinyAES / SPECK	Symmetric Permutation	128 bits	O(1)	YES ✓	Device Layer

** O(1) server-side search via symmetric key indexing. All classical schemes broken by Shor's algorithm on CRQC.*

KEY GAP: No deployed framework integrates quantum resistance + lightweight IoMT auth + privacy-preserving AI + formal verification in a single architecture.

Research Objectives

Five measurable goals of the SHIP framework

01 AUTH

LAAP Lightweight Authentication

Design Curve25519 ECDH protocol achieving O(1) server search & ≥70% overhead reduction vs. RSA-3072 on IoMT MCUs

02 PQC

Post-Quantum Cryptography Integration

Integrate NIST FIPS 203 (ML-KEM) + FIPS 204 (ML-DSA) for WAN security; mitigate payload inflation via TCP initcwnd tuning

03 DP

Differential Privacy Federated Learning

Deploy DP-FL IDS at gateway layer; ≥95% accuracy at $\varepsilon=2.0$, $\delta=10^{-5}$ using Gaussian noise mechanism with FedAvg

04 ABE

CP-ABE Access Control (HIPAA/GDPR)

Implement LSSS-based CP-ABE for hybrid IPFS/Blockchain storage; GDPR Art.17 via key revocation; HIPAA minimum-necessary

05 LTL

Formal Smart Contract Verification

Verify healthcare smart contract safety & liveness via Linear Temporal Logic (LTL) model checking using SPIN/NuSMV

Proposed System Architecture: SHIP Framework

Secure Hybrid IoMT Protocol — Four-Tier Cryptographic Stack

LAYER 1

Device Layer (IoMT Perception)

LAAP Curve25519 ECDH · AES-128-GCM · Constrained MCUs

Wearables · Implantables · ECG Patches · Glucose Sensors

LAYER 2

Gateway Layer (Edge Compute)

DP-FL (FedAvg) · CP-ABE Encrypt · IPFS Upload · ARM Cortex-A

Hospital Edge Nodes · Clinic Gateways · IoMT Aggregators

LAYER 3

Authority Layer (Health Authority)

ML-KEM-768 + ML-DSA-65 · PQC TLS 1.3 · FL Aggregator · IPFS Cluster

Key Mgmt (CP-ABE) · Model Aggregator · Compliance Engine

LAYER 4

Smart Contract Layer (Blockchain)

LTL-Verified Contract · On-Chain CID Registry · RBAC · Audit Log

Consensus Nodes · Immutable Ledger · Zero PHI On-Chain

HIC
LAY
ER
NG
PRI
NCI
PLE

Authentication Layer: LAAP Protocol

Lightweight Adaptive Authentication Protocol on Curve25519

Curve25519 ECDH

Montgomery curve with ladder arithmetic — constant-time, side-channel resistant scalar multiplication

O(1) Device Lookup

Symmetric-cipher indexed registry; server search scales to 10,000+ devices with zero latency penalty

Perfect Forward Secrecy

Fresh ephemeral key pair (k_i, k_g) per session — past sessions secure even if long-term keys are later compromised

Session Key Derivation

K_sess = HKDF-SHA256(S, 'SHIP-v1.0', ID_i ‖ ID_g) — authenticated with AES-128-GCM

DEVICE
Generate ephemeral pair
(k_i_eph, R_i = k_i·G)

→ GATEWAY
Send {ID_i, R_i, HMAC(ts)}

GATEWAY
O(1) lookup Q_i;
ECDH: S = k_g · R_i

← DEVICE
K_sess = HKDF(S,...)
Mutual MAC verify ✓

SESSION
AES-128-GCM · 92.22 ms · PFS ✓

92.22 ms · 71.96% faster than RSA-3072 · O(1) server search

Post-Quantum Cryptography Layer

NIST FIPS 203 (ML-KEM) & FIPS 204 (ML-DSA) — August 2024 Standards

Mathematical Foundation: Module Learning With Errors (M-LWE) over polynomial ring $R_q = Z_q[X]/(X^n+1)$, n=256, q=3329

ML-KEM (FIPS 203)

CRYSTALS-Kyber · Key Encapsulation Mechanism

Variant	Pubkey	Ciphertext	NIST Level
ML-KEM-512	800 B	768 B	Level 1
ML-KEM-768*	**1184 B**	**1088 B**	**Level 3**
ML-KEM-1024	1568 B	1568 B	Level 5

** SHIP uses ML-KEM-768 on WAN*

ML-DSA (FIPS 204)

CRYSTALS-Dilithium · Digital Signature Algorithm

Variant	Pubkey	Signature	NIST Level
ML-DSA-44	1312 B	2420 B	Level 2
ML-DSA-65*	**1952 B**	**3309 B**	**Level 3**
ML-DSA-87	2592 B	4627 B	Level 5

** SHIP uses ML-DSA-65 on WAN*

⚠ Payload Inflation Challenge

ECDH classical: ~64 bytes total handshake
ML-KEM-768 + ML-DSA-65: ~6,000+ bytes
Exceeds Ethernet MTU → TCP fragmentation → +2.4 RTT
BLE limit: 244 bytes — PQC not viable on radio layer

✓ SHIP Mitigation Strategy

TCP initcwnd = 20 × MSS (gateway NIC config)
Full PQC handshake in 1 congestion window on LAN
UDP fragmentation logic for CoAP/DTLS edge links
Hybrid mode: X25519 + ML-KEM-768 (crypto-agility)

Privacy & Access Control Layer

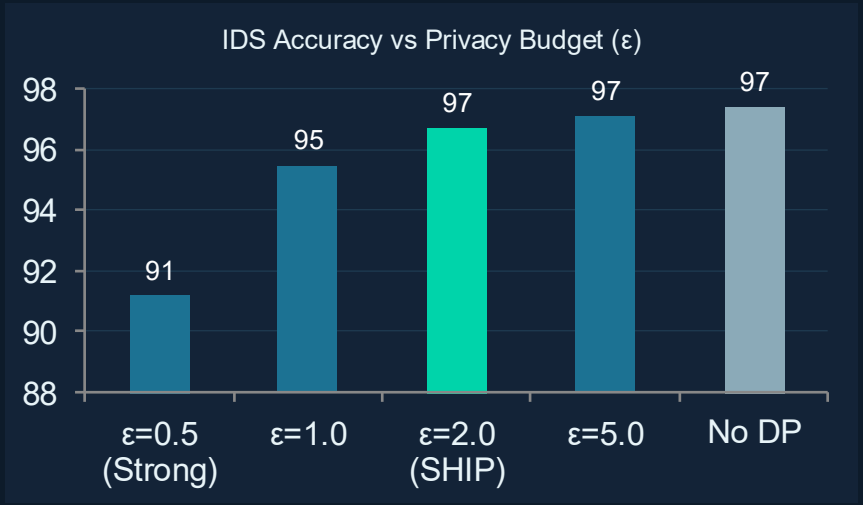
DP-FL Intrusion Detection + CP-ABE Fine-Grained Access Control

DP-FL · Differential Privacy Federated Learning

Gaussian Mechanism:

$$\Delta\tilde{W}_j = \text{clip}(\Delta W_j, C) + N(0, \sigma^2 \cdot C^2 \cdot I)$$

$\sigma = \sqrt{(2 \cdot \ln(1.25/\delta))} \cdot \Delta f / \varepsilon$



CP-ABE · Ciphertext-Policy Attribute-Based Encryption

Access Policy:

Encoded in ciphertext via Boolean formula over user attributes (role ∧ hospital ∧ clearance)

LSSS Matrix:

Policy compiled to (M, ρ): shares $\lambda_i = M_i \cdot v$, secret s reconstructed only if attrs satisfy policy

Encryption:

DEK encrypted under policy; AES-256-GCM encrypts payload; CP-ABE encrypts DEK

GDPR Art. 17:

RTBF → key revocation → IPFS delete → on-chain CID orphaned = cryptographic destruction

HIPAA:

LSSS share threshold ensures 'minimum necessary' access; no over-provisioned decryption keys

System Workflow: End-to-End SHIP Operations

Four-phase cryptographic pipeline from device to blockchain

01 Secure Init & Auth

1. IoMT: generate ephemeral Curve25519 key pair
2. Gateway: O(1) lookup; ECDH → K_sess
3. HKDF derive session key; Mutual MAC verify
4. AES-128-GCM session 92.22ms · PFS ✓

02 Threat Detection

1. Gateway LSTM trains on local traffic features
2. Compute gradient ΔW_j via backpropagation
3. Clip + inject DP noise $\Delta\tilde{W}_j = \text{clip} + N(0,\sigma^2C^2I)$
4. FedAvg aggregation 95.48% accuracy ✓

03 Off-Chain Storage

1. Generate DEK; AES-256-GCM payload
2. CP-ABE encapsulate DEK under access policy
3. Upload ciphertext to IPFS cluster
4. Receive CID = SHA-256(content) ✓

04 PQC Blockchain

1. Format tx: {CID, policy hash, metadata}
2. TLS 1.3 + ML-KEM-768 + ML-DSA-65 auth
3. initcwnd = 20×MSS < 50ms PQC latency
4. LTL contract validates → CID committed ✓

Results & Performance Analysis

Quantified improvements across all four cryptographic components

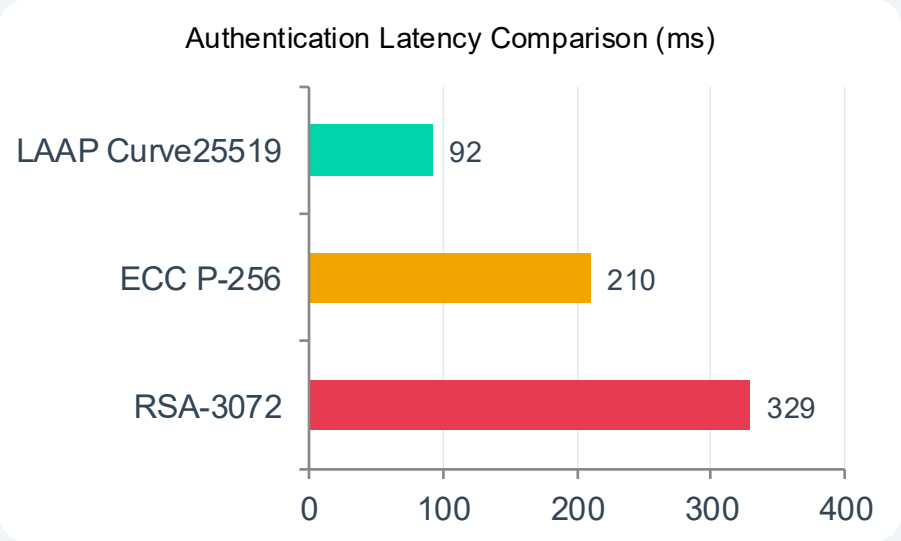
92.22 ms
LAAP Auth Latency
vs 329ms for RSA

71.96%
Overhead Reduction
vs RSA-3072 baseline

95.48%
DP-FL IDS Accuracy
at ε=2.0, δ=1e-5

<50 ms
PQC Latency Penalty
post initcwnd tuning

0
LTL Counterexamples
All 5 invariants verified



Metric	Classical	SHIP
Auth Latency	329 ms (RSA)	**92.22 ms ✓**
Overhead vs RSA	Baseline	**–71.96% ✓**
Server Lookup	O(n) linear	**O(1) constant ✓**
IDS Accuracy	~70% (central)	**95.48% DP-FL ✓**
Quantum Safe	NONE	**Full (M-LWE) ✓**
PQC Handshake	N/A	**~50ms (tuned) ✓**
Smart Contract	Unverified	**LTL Proven ✓**
GDPR Art. 17	Not possible	**Key Revoke ✓**

Conclusion & Regulatory Compliance

SHIP achieves all objectives — fully HIPAA & GDPR-compliant

LAAP Authentication

92.22 ms · 71.96% reduction · O(1) server search · PFS guaranteed · Curve25519 side-channel resistant

Post-Quantum Security

FIPS 203 ML-KEM + FIPS 204 ML-DSA · M-LWE hardness · Quantum-resistant WAN · Hybrid crypto-agility mode

Privacy-Preserving AI

DP-FedAvg · ε=2.0 · 95.48% IDS accuracy · (ε,δ)-DP formal guarantee · No raw PHI ever leaves gateway

HIPAA Compliance

- Minimum Necessary: LSSS attribute threshold — mathematical enforcement
- Zero PHI on-chain: only CID hashes & policy metadata
- CP-ABE enforces role-based data access at cryptographic level
- Audit completeness: LTL liveness invariant G(access → F audit_logged)

GDPR Article 17 — Right to Be Forgotten

- RTBF Request → Smart contract triggers key revocation event
- IPFS record permanently deleted from distributed storage
- CP-ABE decryption keys cryptographically revoked — data inaccessible
- On-chain CID becomes orphaned hash: anonymous audit trail preserved

SHIP is the first unified framework combining lightweight IoMT auth · quantum resistance · DP-FL privacy · LTL formal verification · full HIPAA/GDPR compliance

Future Scope & Research Roadmap

Five key directions for real-world clinical deployment

Near Term (1–2 yrs)

PQC on IoMT Hardware

ARMv9.5-A PQC acceleration → ML-KEM on Cortex-M33; direct device-layer quantum resistance without gateway relay

Near Term (1–2 yrs)

Hybrid KEM Deployment

IETF RFC 9420 hybrid: X25519 + ML-KEM-768 → crypto-agility; hedge against both quantum realization & lattice cryptanalysis

Mid Term (2–4 yrs)

Byzantine-Resilient FL

Krum & Bulyan aggregation protocols for adversarial gateway nodes; protect against poisoning & model inversion attacks

Mid Term (2–4 yrs)

IEEE 11073 / HL7 FHIR

Standardize SHIP within IEEE 11073 PHD & HL7 FHIR interoperability specifications; prerequisite for hospital regulatory approval

Long Term (4+ yrs)

ZK-Proof Identity Layer

Integrate SNARK-based zero-knowledge proofs for consent & identity verification; full attribute privacy beyond CP-ABE scope

Key References

IEEE Format Citations

[1] Y. Zhu et al., "LAAP: Lightweight Anonymous Authentication Protocol for IoT," Wireless Comm. & Mobile Computing, 2022. doi:10.1155/2022/8768928

[2] NIST, "FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM)," August 2024.

[3] NIST, "FIPS 204: Module-Lattice-Based Digital Signature Algorithm (ML-DSA)," August 2024.

[4] M. Kannwischer et al., "Benchmarking ML-KEM and ML-DSA on ARM Cortex-M0+," arXiv:2603.19340v4, 2025.

[5] M. Anwar & M. Hoque, "Optimizing Privacy-Accuracy Trade-Off in IoT IDS with DP," Semantic Scholar, 2024.

[6] C. Dwork & A. Roth, "The Algorithmic Foundations of Differential Privacy," Foundations & Trends TCS, vol.9, 2014.

[7] Bethencourt, Sahai & Waters, "Ciphertext-Policy Attribute-Based Encryption," IEEE S&P, 2007.

[8] NIST, "NIST Releases First 3 Finalized Post-Quantum Encryption Standards," nist.gov, August 2024.

[9] NIST CSRC, "Impact of Data-Heavy PQC on TLS 1.3 TTLB," 5th PQC Standardization Conference, 2024.

[10] PMC, "Advancing Compliance with HIPAA & GDPR in Healthcare: A Blockchain Strategy," 2025. PMC12563691

Thank You

Questions & Discussion

LAAP	PQC	DP-FL	CP-ABE	Formal V.
Curve25519 ECDH · O(1) complexity · PFS mechanism	ML-KEM/ML-DSA lattice math · FIPS 203/204 · payload mitigation	Gaussian noise calibration · ε-δ privacy · FedAvg aggregation	LSSS matrix · bilinear pairings · GDPR key revocation	LTL invariants · SPIN model checking · safety & liveness proofs

[Student Name] · [Roll No.] · MUJ CSE · Jan–May 2026